

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2016

Mathematics

Paper 2
Ordinary Level

Time: 2 hours

300 marks

School stamp

Running total	
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For examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3		13	
4		14	
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 14 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if all necessary work is not clearly shown.

You may lose marks if the appropriate units of measurement are not included, where relevant.

You may lose marks if your answers are not given in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1

(Suggested maximum time: 10 minutes)

- (a) Choose the most appropriate unit from the list below to complete the following sentences.

litres	millimetres	kilometres	metres	grammes
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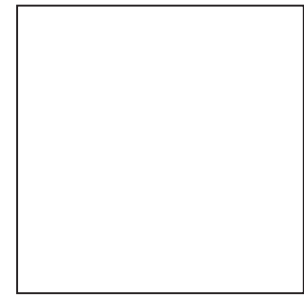
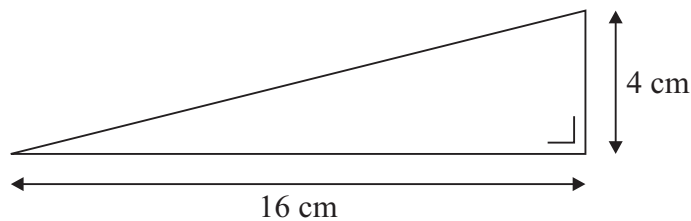
- (i) The distance between Navan and Dublin is about 55

- (ii) The capacity of a carton of milk is 2

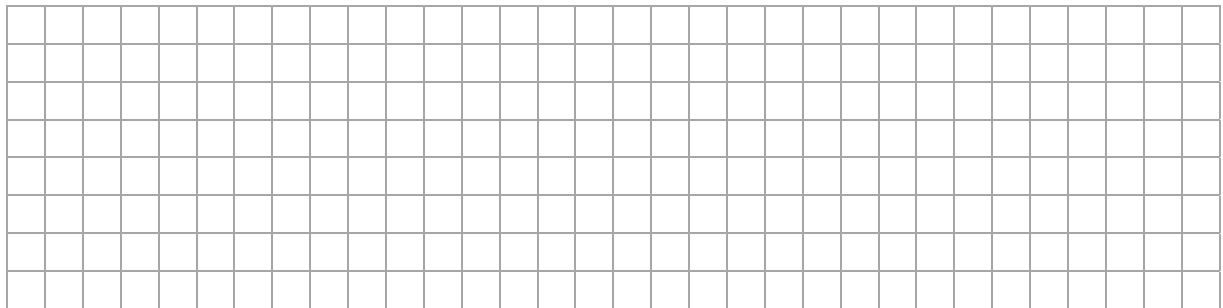
- (iii) The weight of an orange is about 130

- (iv) The height of a garden shed is approximately 2.5

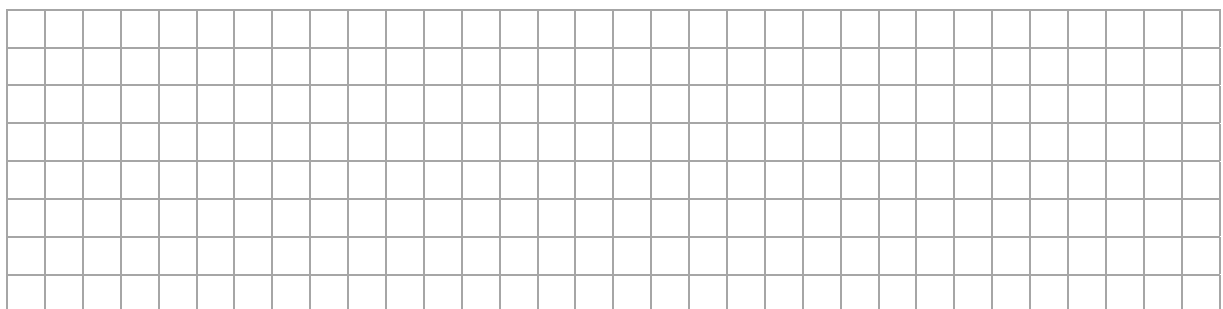
- (b) The diagram below shows a right-angled triangle and a square.



- (i) Find the **area** of the triangle.



- (ii) The area of the square is twice the area of the triangle.
Find the **length** of each side of the square.



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Question 2

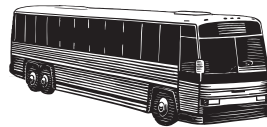
(Suggested maximum time: 10 minutes)

- (a)** An extract from a bus timetable is shown below.
Use the timetable to answer the following questions.

Dublin (departs)	07:45	11:00	14:00	17:30	20:00	21:00
Longford	09:57	13:12	16:12	19:27	22:12	23:12
Carrick-on-Shannon	10:40	13:55	16:55	20:15	22:50	23:50
Sligo (arrives)	11:33	14:48	17:53	21:03	23:40	00:35

A bus leaves Dublin for Sligo at 07:45.

- (i)** At what time does the bus arrive in Sligo?

[illegible]

- (ii) How long does the journey take? Give your answer in hours and minutes.

[illegible]

Pat arrives at the Longford stop at 7:09 p.m. He waits for the next bus to Sligo.

- (iii)** How long does he have to wait for the next bus to arrive?

[illegible]

- (iv) The bus journey from Longford to Sligo is delayed by roadworks for 25 minutes. However, the bus driver reduces this delay by 8 minutes due to light traffic. How long does Pat's journey take?

[illegible]

- (b)** Tim drove from Dublin to Sligo, a distance of 210 km.
It took him 2 hours and 30 minutes.
Hannah also drove from Dublin to Sligo on the same road.
She drove at an average speed of 85 km/h.



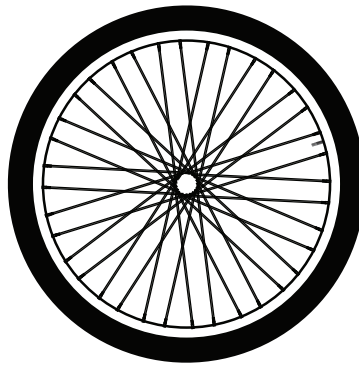
Which driver had the greater average speed? Give a reason for your answer.

[illegible]

Question 3

(Suggested maximum time: 5 minutes)

Both wheels on Jamie's bicycle have a diameter of 62 cm.



- (a) Find the **radius** of each wheel.

[illegible]

- (b)** Find the **circumference** of each wheel. Use $\pi = 3.142$.

Jamie cycles his bicycle in a straight line in the school yard. The front wheel makes 15 complete revolutions.

- (c) How far does the bicycle travel?
Give your answer correct to the nearest metre.

[illegible]

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Question 4

(Suggested maximum time: 15 minutes)

Peter wishes to investigate the healthy living styles of people in his local area. He plans to carry out a survey to find out how much sport and exercise people do.



- (a) Peter wishes to ask people if they are members of a gym.

Which two words below describe the type of data collected from the answers to this question in the survey?

Categorical	Numerical	Ordinal	Nominal	Continuous	Discrete
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[illegible]

- (b)** Peter wishes to find out how much time gym members spend in the gym each week. He is considering the following question for the survey.

Q. How much time do you spend in the gym?

☐ 0–1 hours ☐ 1–2 hours ☐ 3–4 hours

- (i) Explain why this is **not** a suitable question.

[illegible]

- (ii)** Rewrite the question in a suitable form.

[illegible]

- (c) Peter carries out his survey by making the questionnaire available online. Explain one advantage and one disadvantage of using this method to collect data.

[illegible]

- (d) The ages of 20 people who take part in Peter's survey are shown below.

21	34	16	35	46
63	23	29	57	26
24	34	17	28	50
19	42	38	20	33

- (i) Represent the data on a stem-and-leaf diagram.

1																			
2																			
3																			
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Key: 1 6 =																			

- (ii) How many people under 25 years of age take part in the survey?

- (iii) Find the median age.

- (iv) The next person who takes part in the survey is 15 years of age.
Put a tick (✓) in the correct box to show how this will affect the range.

Range will decrease ☐	Range will increase ☐	Range will stay the same ☐
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Give a reason for your answer.

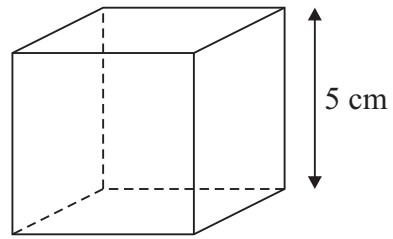
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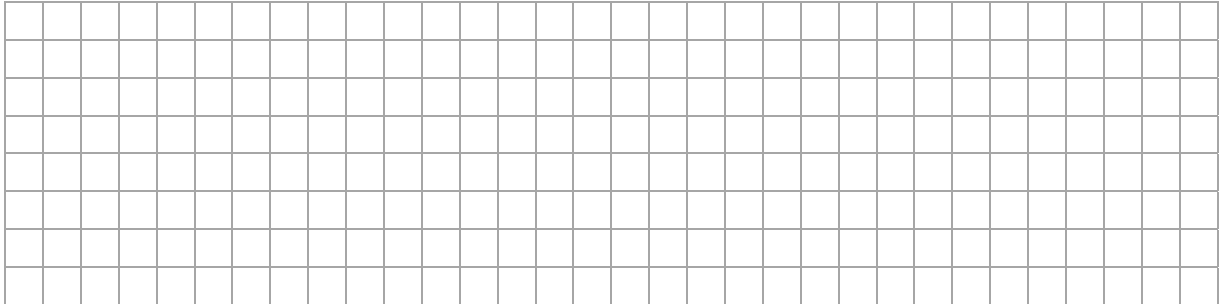
Question 5

(Suggested maximum time: 5 minutes)

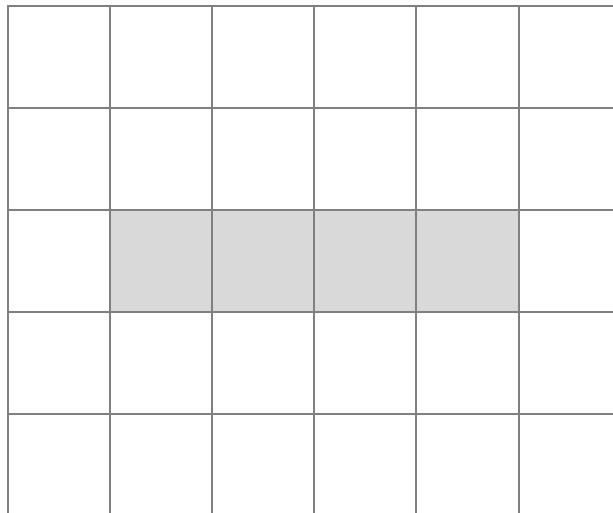
A cube has sides of length 5 cm.



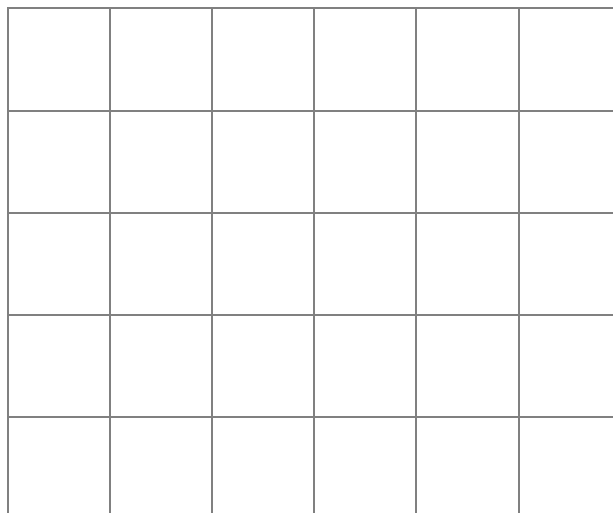
- (a) Find the total **surface area** of the cube.



- (b) Complete the **net** of this cube in the diagram below by shading in two squares.



- (c) Draw an alternative net of a cube to the one you have completed in part (b).



Question 6 (Suggested maximum time: 10 minutes)

Question 6 (Suggested maximum time: 10 minutes)

- (a) Five cards, arranged in order from smallest to largest, are shown below. Each card has a natural number assigned to it. The first card contains the number 5, as shown.

5				
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The **range** of numbers on the cards is 6.
The **median** of the numbers on the cards is 6.
The **mode** of the numbers on the cards is 5.
The **mean** of the numbers on the cards is 7.

- (i) Use this information to find the missing numbers on the four other cards. Write your answers on the cards above.

A large grid of 20 columns and 10 rows, intended for drawing. The grid is composed of thin black lines forming a series of small squares.

- (ii) Jim picks a card at random.
What is the probability that the card has the number 5 on it?

[illegible]

Answer =

- (b) Another five cards are shown below.

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Write on the cards five numbers for which the following statements are true.

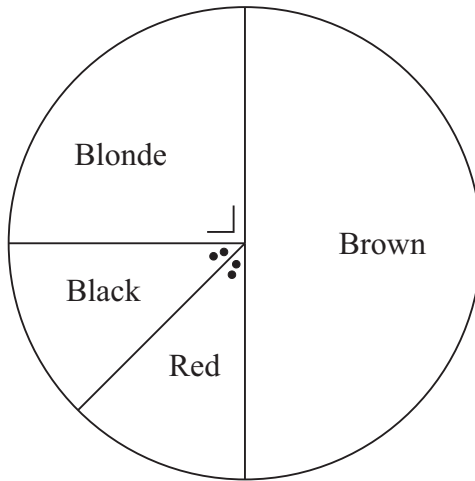
The **mode** of the numbers on the cards is 8.
The **mean** of the numbers on the cards is 7.

[illegible]

Question 7

(Suggested maximum time: 5 minutes)

In a survey, a group of students were asked to indicate their hair colour. The results are displayed in the pie chart below.



- (a)** Four events, **A**, **B**, **C** and **D**, are listed below.

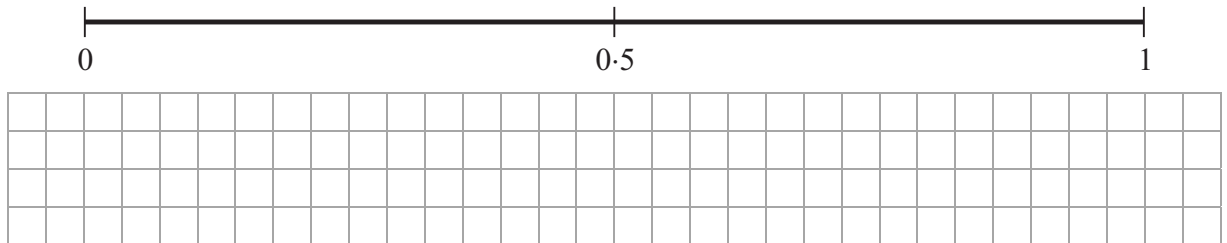
A: A student, chosen at random, has **brown** hair.

B: A student, chosen at random, has **grey** hair.

C: A student, chosen at random, has **blonde** hair.

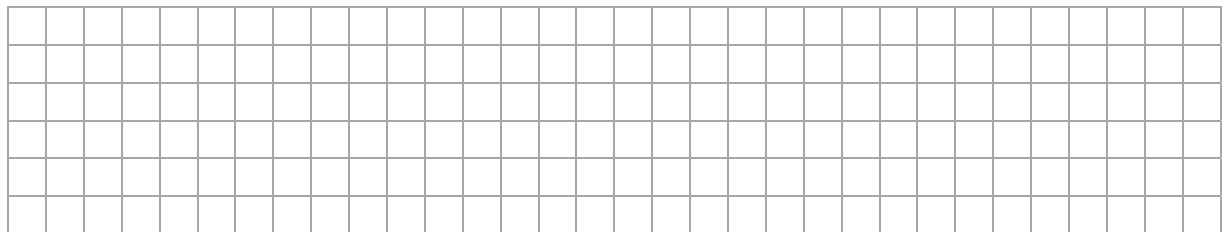
D: A student, chosen at random, has **red**, **brown** or **black** hair.

Place each of the letters **A**, **B**, **C** and **D** in the correct position on the probability scale below, to show the probability of each event.

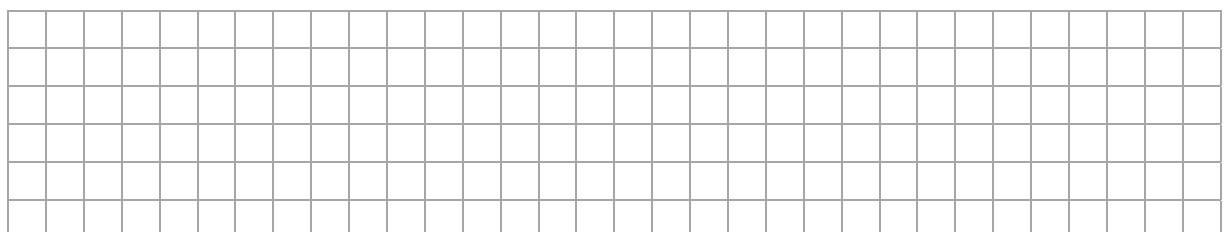


- (b)** Six students from the survey have **black** hair.

- (i) How many students have **brown** hair?



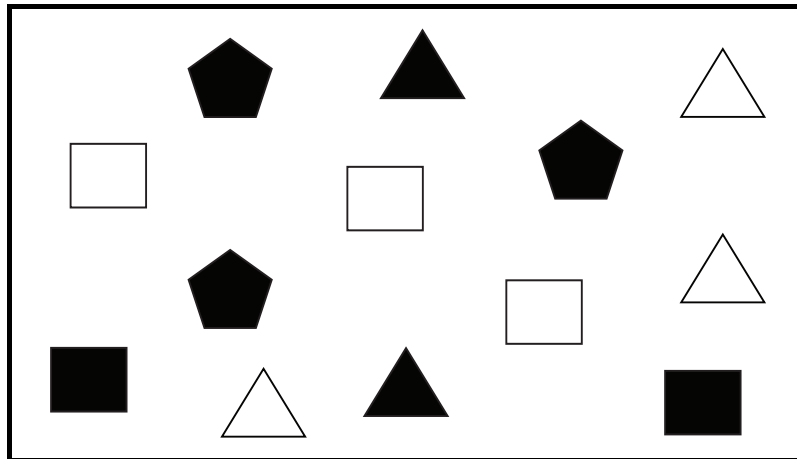
- (ii)** Find the total number of students who took part in the survey.



Question 8

(Suggested maximum time: 5 minutes)

A box contains some 3-sided, 4-sided, and 5-sided shapes. The shapes are either black or white.



- (a)** Complete the table below to show the number of different types of shapes in the box.

	Black	White	Total
3-sided shape		3	5
4-sided shape	2		
5-sided shape		0	
Total			13

- (b)** Amy picks a shape at random from the box.

- (i)** Find the **probability** that this shape is **white** and **3-sided**.

[illegible]

Answer =

- (ii) Find the **probability** that this shape is **4-sided**.

[illegible]

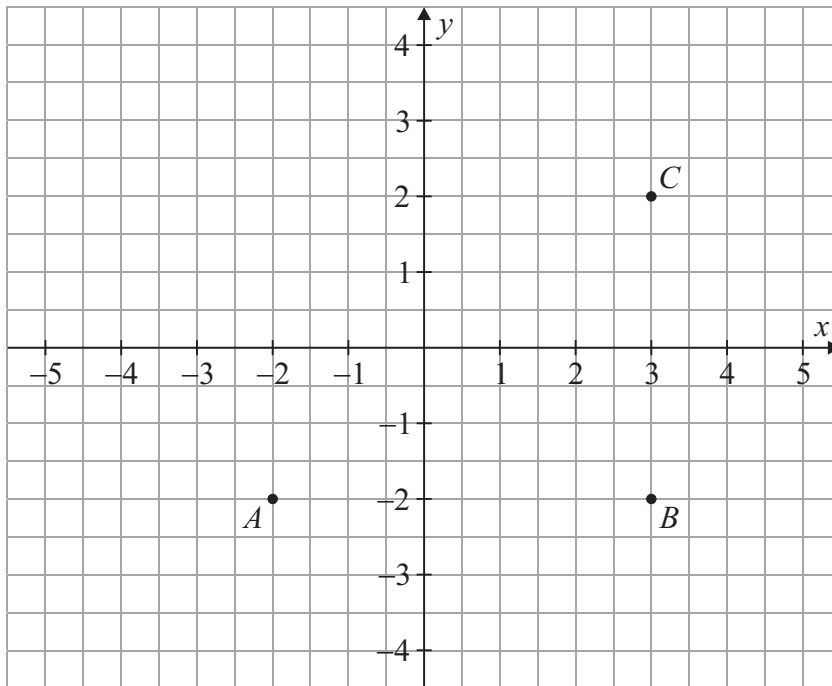
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Question 9

(Suggested maximum time: 10 minutes)

The points A , B and C are shown on the co-ordinate grid below.



- (a)** Write down the co-ordinates of the points A , B and C .

$$A = \left(\begin{array}{cc} & \\ & \end{array} \right)$$

$$B = \begin{pmatrix} & \\ & \end{pmatrix}$$

$$C = \begin{pmatrix} & \\ & \end{pmatrix}$$

- (b)** $ABCD$ is a rectangle. Plot the point D on the co-ordinate grid above and hence, write down the co-ordinates of the point D .

$$D = \begin{pmatrix} & \\ & \end{pmatrix}$$

- (c) Using your **ruler**, draw a diagonal of the rectangle $ABCD$ on the co-ordinate grid.

- (d) David says the slope of a diagonal of the rectangle $ABCD$ is $\frac{4}{5}$.

Joanne says the slope of a diagonal of the rectangle $ABCD$ is $-\frac{4}{5}$.

Put a tick (✓) in the correct box to show which statement is true.

Only David is correct ☐	Only Joanne is correct ☐	Both are correct ☐
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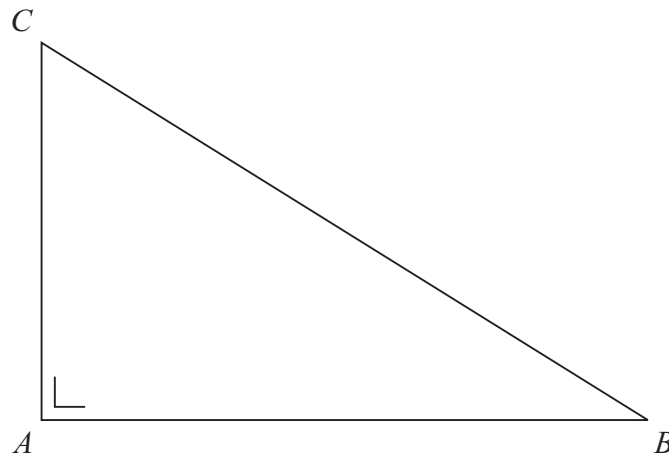
Give a reason for your answer.

Reason:

Question 10

(Suggested maximum time: 10 minutes)

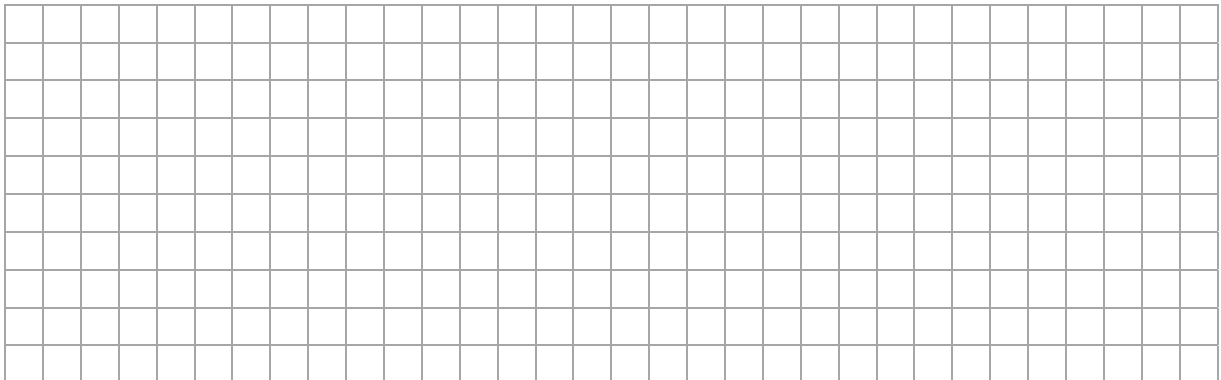
The diagram below shows a right-angled triangle ABC .



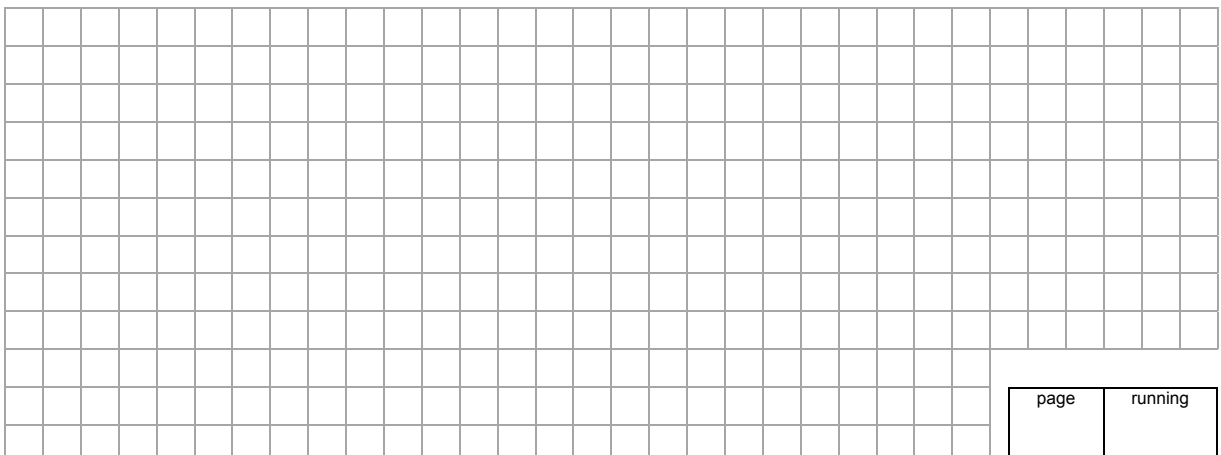
- (a) Using your **ruler**, find the length of $[AB]$ and the length of $[AC]$.
Give your answers correct to the nearest centimetre.

$$|AB| = \boxed{} \text{ cm} \qquad |AC| = \boxed{} \text{ cm}$$

- (b) Bobby says the length of $[BC]$ is $\sqrt{89}$ cm.
Use the **Theorem of Pythagoras** to show that he is correct.



- (c) Bobby uses his **protractor** to find $|\angle ABC| = 32^\circ$, correct to the nearest degree.
Eva lost her protractor but says that she can find $|\angle ABC|$ using trigonometric ratios.
Show how Eva can get the same answer as Bobby without using her protractor.

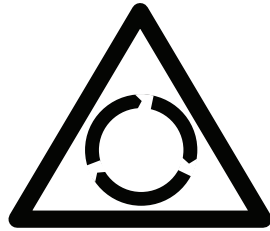


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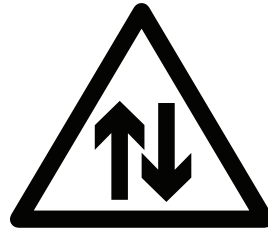
Question 11

(Suggested maximum time: 5 minutes)

The diagram below shows four road signs.



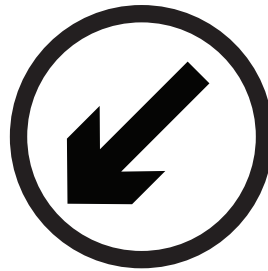
A



B



C



D

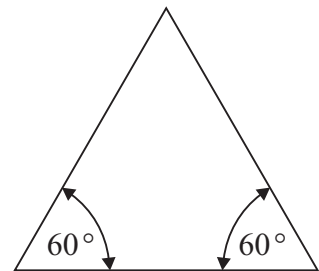
- (a)** Two of the road signs are in the shape of a triangle, as shown.

Paul says the two triangles are **isosceles**.

Liam says the two triangles are **scalene**.

Claire says the two triangles are **equilateral**.

Who is correct? Give a reason for your answer.

[illegible]

- (b)** Which two of these road signs have exactly one axis of symmetry?

Answer:

and

- (c) One of these road signs has rotational symmetry. Which road sign is it?

Answer:

Page 10 of 10

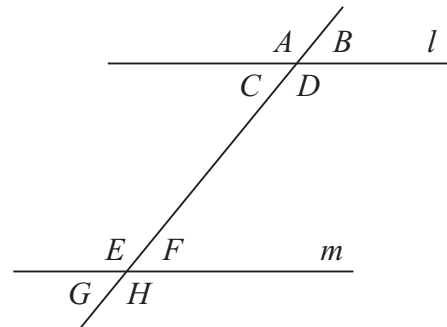
Question 12

(Suggested maximum time: 10 minutes)

- (a) Using only a **compass** and a **straight-edge**, construct the perpendicular bisector of the line segment $[XY]$. Show all construction lines clearly.



- (b) In the diagram below, line l is parallel to line m .



- (i) Which angle is **vertically opposite** to angle H ?

Answer:

- (ii) Which angle is **corresponding** to angle E ?

Answer:

- (iii) Which angle is **alternate** to angle C ?

Answer:

- (iv) Which angle is **corresponding** to angle F ?

Answer:

- (v) Which angle is **alternate** to angle D ?

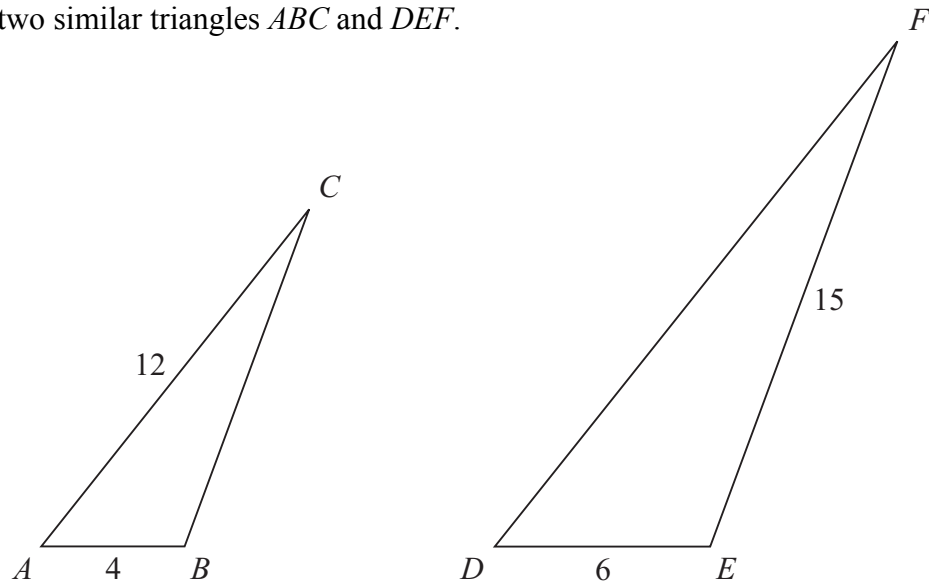
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Question 13

(Suggested maximum time: 5 minutes)

The diagram shows two similar triangles ABC and DEF .



- (a) Write down **one** property of similar triangles.

[illegible]

- (b) Find the length of $[DF]$.

A blank sheet of graph paper featuring a uniform grid of small squares. The grid consists of 20 columns and 10 rows, providing a structured space for drawing or writing.

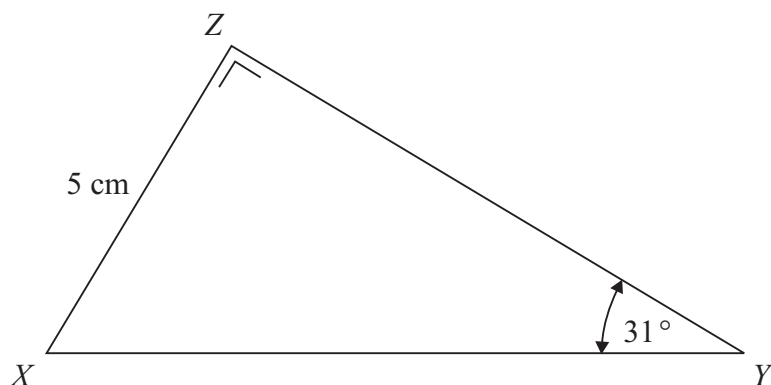
- (c) Find the length of $[BC]$.

A large grid of graph paper with 20 columns and 10 rows. The grid is composed of small squares, with a slightly larger square at the top left corner, likely for a title or header. The grid is empty and ready for use.

Question 14

(Suggested maximum time: 5 minutes)

XYZ is a right-angled triangle with $|XZ| = 5$ cm and $|\angle XYZ| = 31^\circ$.



- (a) Find $|\angle ZXY|$.

[illegible]

- (b) Jim says that $[XY]$ is the side **adjacent** to the angle $\angle ZXY$. Is he correct? Give a reason for your answer.

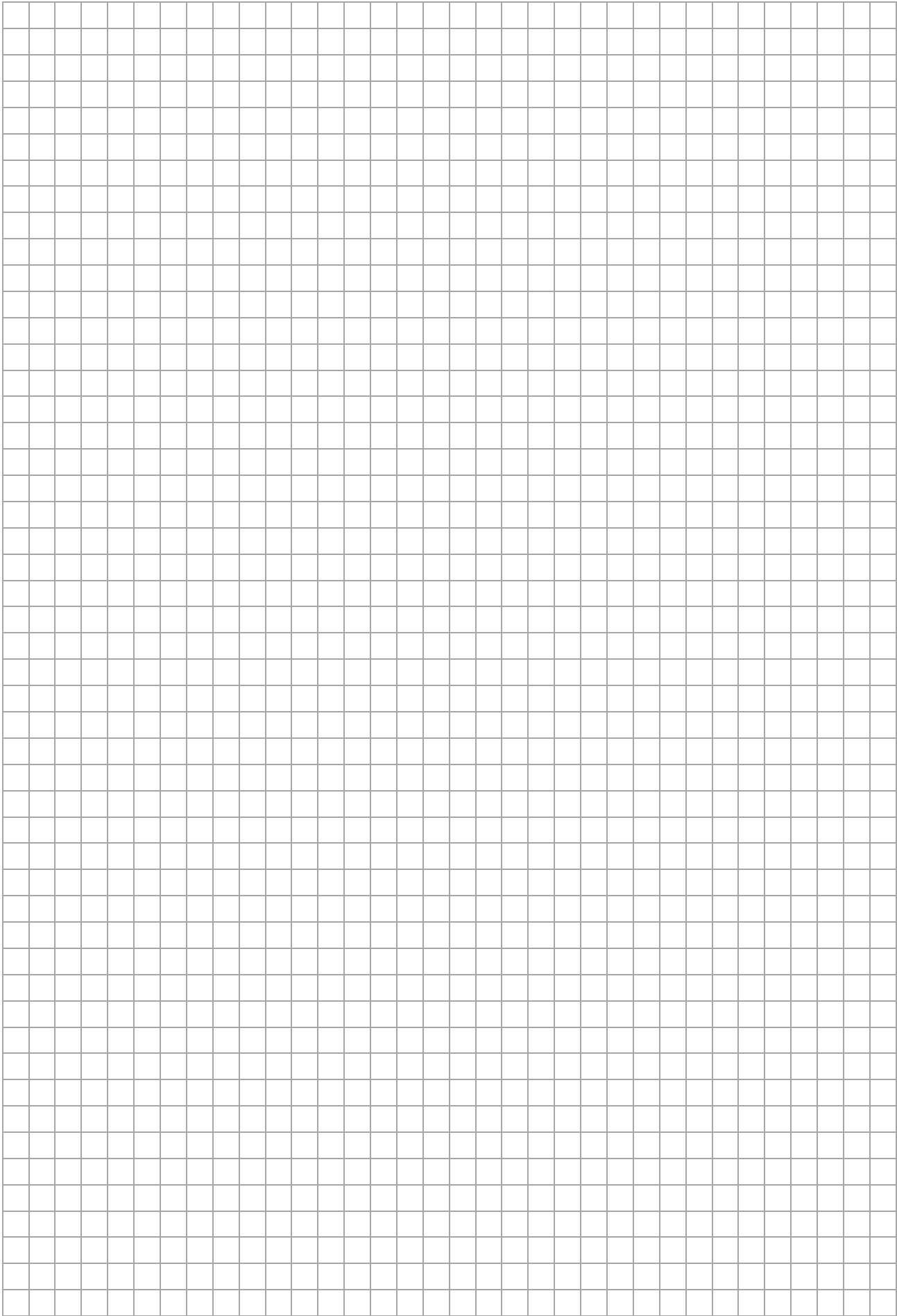
[illegible]

- (c) Find the length of $[XY]$. Give your answer correct to two decimal places.

[illegible]

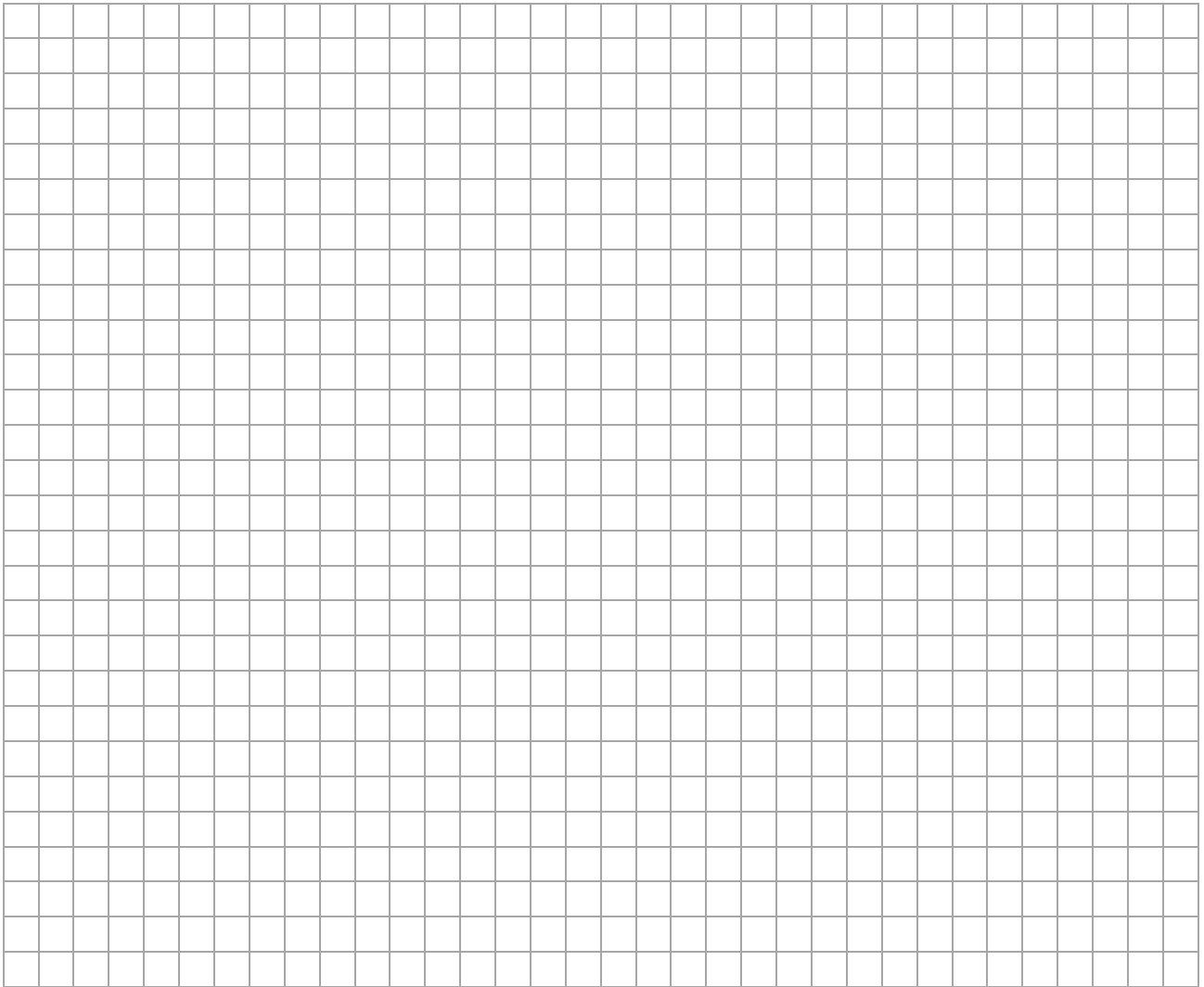
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You may use this page for extra work.



A full-page grid of graph paper. The grid consists of small squares. In the bottom right corner, there is a small table with two columns and two rows. The first row contains the words 'page' and 'running'. The second row is empty.

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Pre-Junior Certificate, 2016 – Ordinary Level

Mathematics – Paper 2

Time: 2 hours

